FitBuddy – Architecture Overview

FitBuddy follows a modular FastAPI-based architecture.

Architecture Components

Layer / Component	Technology / Responsibility
Frontend	HTML + Jinja2 templates for user input and display
Backend	FastAPI for route handling and application logic
AI Layer	Google Gemini APIs (Pro & Flash)
Database	SQLite with SQLAlchemy for persistence

Frontend Responsibilities

Users interact via:

- **index.html** – Input form
- **result.html** – Plan, tip, feedback
- **all_users.html** – Admin panel

Backend Responsibilities

- Receives form data
- Calls Gemini models
- Stores user information and plans
- Renders dynamic templates

AI Integration Points

Function	AI Model
Workout generation & feedback updates	Gemini Pro
Nutrition tips	Gemini Flash